Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **C++ systems**, **cross-platform integrations**, concurrent services, and automated validation workflows; grounded in **computer science fundamentals**, debugging, OOP, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Restructured **fault-tolerant asynchronous orchestration** with explicit job states and failure isolation, processing 1K+ tasks while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.
- Architected multi-tenant application infrastructure for 500+ users, reducing indexed-query latency from 1.2 seconds to 350 milliseconds through schema design, access-pattern optimization, and disciplined production performance analysis workflows.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed **concurrent processing** with multithreading and secure ingestion, evaluating 10K+ text and handwritten submissions daily through recoverable execution paths while maintaining 90%+ grading consistency in production environments.
- Built React interfaces and Python services for uploads, configuration, history, and live processing feedback, reducing user turnaround 60% while supporting reliable production workflows used by 150+ instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, California*

- Built a PyTorch **simulation pipeline** processing 100K+ trajectory logs, improving long-horizon planning 18% through reproducible feature transformations, benchmark automation, regression testing, and detailed behavioral failure analysis workflows.
- Developed a spatial-temporal encoder combining map geometry, historical trajectories, and agent context, improving closed-loop stability 15% across complex scenarios through controlled evaluation and systematic root-cause debugging workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Engineered **cross-platform software** with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments reliably.
- Troubleshot object-oriented API integrations and AWS data flows, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable modules and structured reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight microservices spanning authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across distributed production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ successful validation and cutting release time 50% through repeatable configurations, staged testing, and monitoring.

Skills

Languages: C++, Python, Java, TypeScript, JavaScript, SQL, Bash
Systems: OOP, Data Structures, Algorithms, Operating Systems, Concurrency, Multithreading, Networking, Distributed Systems, State Machines
Testing & Tools: Unit Testing, Integration Testing, Regression Testing, Fault Injection, Debugging, Performance Analysis, PyTest, Mockito, Git, CMake, Docker, Jenkins, CI/CD
Platforms: Linux, Windows, macOS, AWS, Azure, REST APIs, WebSockets, Microservices

Projects

Vehicle Integration Test Harness | *C++17, CMake, State Machines, Automated Testing*

- Implemented a C++17 matrix-headlight controller simulator with versioned frames, checksums, telemetry, and explicit states; authored nine deterministic assertions for corrupted input, stale sequencing, timeout recovery, and safe-state behavior.

CPU Scheduling Simulator Live | *C++, React, Operating Systems*

- Implemented six preemptive and non-preemptive scheduling algorithms with React controls, real-time Gantt visualization, and dynamic latency metrics across 1K+ process combinations, applying systems fundamentals to interactive resource management.

Chat2SQL Code | *Java, TypeScript, Docker, SQL*

- Worked across a Java and TypeScript codebase with Docker, Maven, Node.js, and Yarn, supporting ten database engines and reliable cross-platform workflows on Windows, macOS, and Linux environments.